5G MAG The Media Connectivity Association



OVERVIEW AND UPDATES · CONTENT DELIVERY AND STREAMING

Content Delivery Protocols and Tools

Content delivery origin, packaging, protocols and related tools

IETF 3GPP ETSI

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About the project

The documents the project rests on

RFC 3926

FLUTE

File Delivery over Unidirectional Transport: files from one sender to any number of receivers, built on ALC and LCT.

3GPP TS 26.346

Technical Specification

FLUTE in MBMS download delivery, with the MBMS Download Profile. TS 26.517 adds the Object Distribution Method of 5G MBS.

ETSI TS 103 973

Technical Specification

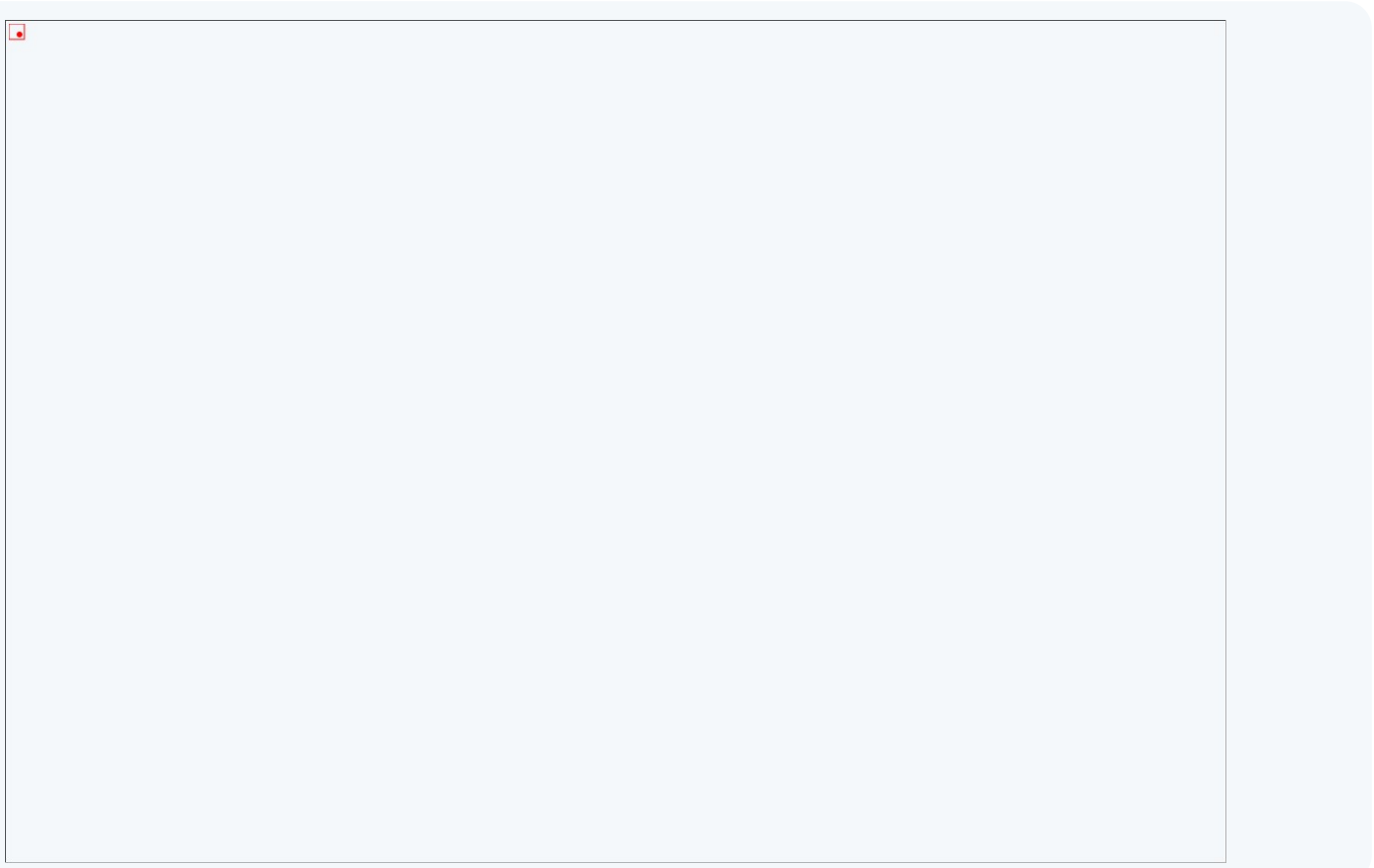
The Coded Multisource Media Format (CMMF), a container for coded media from several sources.

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The building blocks

How FLUTE is built on ALC, and what each block gives it



LCT (Layered Coding Transport)

In-band session management: each packet carries a session and an object identifier.

CC (congestion control)

MBMS download does not use it.

FEC (Forward Error Correction)

Gives ALC its reliability, with the choice of an FEC code.

ALC (Asynchronous Layered Coding)

Combines the three building blocks.

FLUTE

A fully specified protocol to transport files, with a File Delivery Table that indexes the files in band.

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The three FLUTE profiles

FLUTE as the IETF defines it, and the two ways 3GPP profiles it

RFC 3926

Unprofiled FLUTE

FLUTE with no 3GPP constraints, for any FLUTE sender and receiver.

TS 26.346

MBMS download

BM-SC and MBMS Client. The MBMS Download Profile sets the usage of the FDT attributes and elements.

TS 26.517

MBS Object Distribution

MBSTF and MBS Client. Conforms to the MBMS Download Profile, with additional requirements.

Only the MBMS Download Profile has a name in the specifications. Both 3GPP profiles are defined on FLUTE version 1.

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How it grew

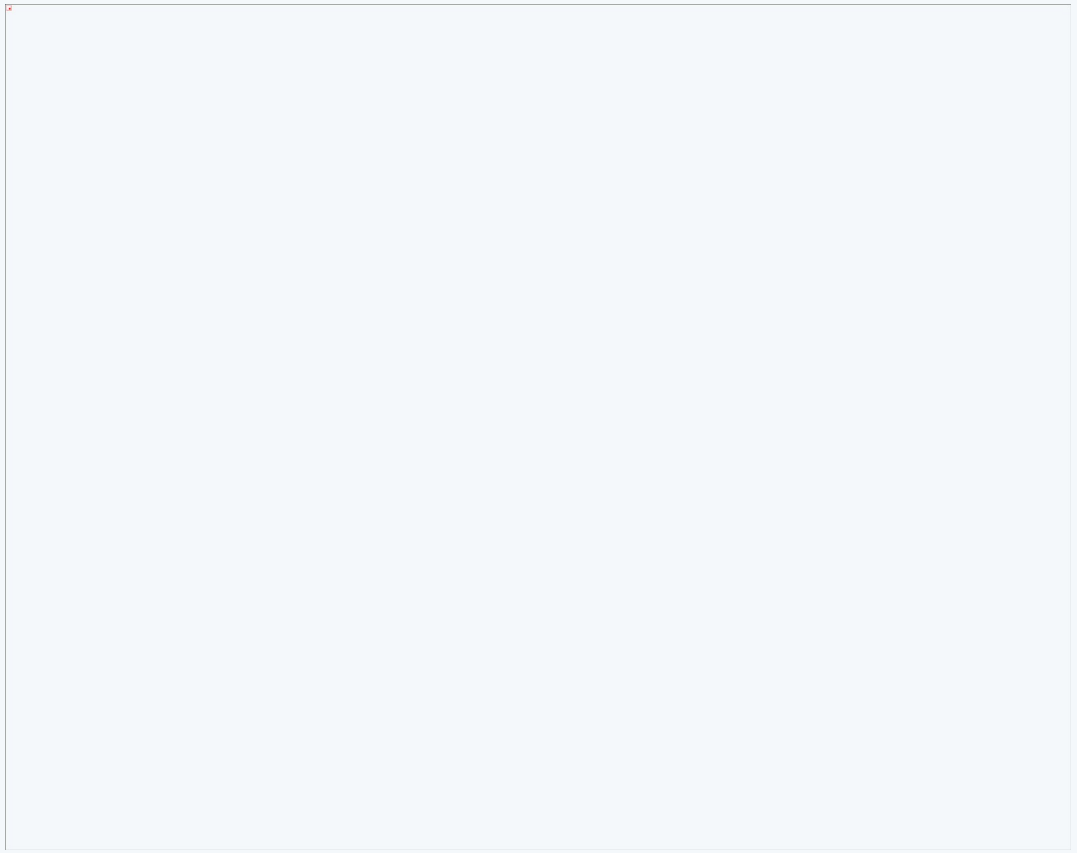
FLUTE, its 3GPP profiles, ROUTE and CMMF over time

2002 to 2004	MBMS	2012 and 2022	Release 17	Release 19 and 20
ALC and LCT in December 2002, then FLUTE version 1 in October 2004, all Experimental RFCs.	From Release 6, TS 26.346 extends the FLUTE File Delivery Table with its own schema extensions, one set for each release that added to them.	FLUTE version 2 (RFC 6726) in 2012, on the Standards Track. ROUTE (RFC 9223) in 2022, an Informational RFC.	The MBMS Download Profile is updated in Release 16. Release 17 adds a feature-reduced FDT schema and creates TS 26.517, with the Object Distribution Method.	In-session unicast file repair in both 3GPP profiles. CMMF: V1.1.1 October 2024, V1.2.2 May 2026. Release 20 (work plan of 28 September 2026): no work item lists TS 26.346 or TS 26.517.

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Architecture

Inside the MBSTF, for the MBS Object Distribution Method



Object ingest

Takes objects from the MBS Application Provider, pulled or pushed.

Object segmentation

Partitions an object into payload units suitable for MBS transmission.

Application Layer FEC

Optional: supports object recovery when some packets are not received.

Packetisation

Places the payload units, and optionally the FEC data, into transmission packets.

Packet scheduling

Schedules the outgoing packet stream according to the target bit rate.

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Beyond the three profiles

Also in the scope of the project

RFC 6726

FLUTE version 2

Obsoletes version 1 and is not backwards compatible with it. Both 3GPP profiles refer to version 1.

RFC 9223

ROUTE

Delivers application objects, including objects with real-time constraints, over unidirectional transport. Aligned with FLUTE.

ETSI TS 103 973

CMMF

A container for coded multisource media, so a client can draw from several sources at once. It carries media of an already packaged format.

TS 26.346 and TS 26.517 do not use ROUTE. CMMF is agnostic to the transport; it specifies content delivery protocol instantiations.

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Repositories

The repositories of the project, from their descriptions

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-libflute	a FLUTE server and client, with a demo transmitter and a demo receiver	FLUTE server, FLUTE client	0.12.4 1 Sep 2026	GitHub
rt-cmmf-encoder	a reference implementation of a CMMF Encoder according to ETSI TS 103 973	No encoder code yet	none yet	GitHub
rt-media-origin	a media origin server and packager for content delivery testing	Media origin	none yet	GitHub
rt-media-android-application	an Android test app and a laptop test rig that check that every delivery method of the media origin arrives on a phone and plays as configured	Test app	none yet	GitHub
flute-conformance-validator	tests a FLUTE implementation against the FLUTE profiles of RFC 3926, TS 26.346 and TS 26.517	FLUTE conformance	none yet	GitHub

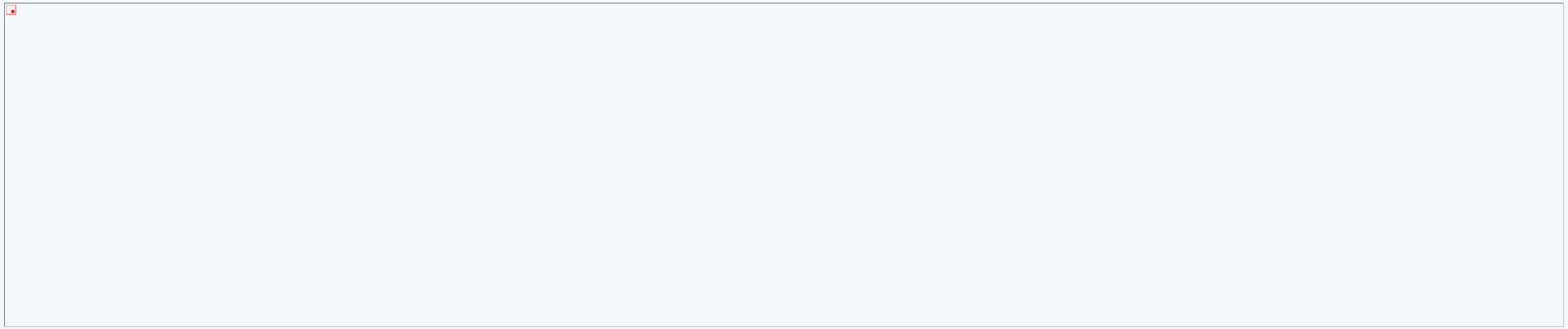
FIND OUT MORE Technology Specifications Reference Tools

● released ● no release yet

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1 **rt-libflute**

Boxes without a badge are not mapped to a repository.

WHERE WE STAND

Content Delivery Protocols and Tools

BROUGHT TO STUDY	ON-GOING	OUTCOMES
	• Implementation of a versatile Media Origin and Packager to support 5G-MAG projects (DASH, HLS, CMAF packaging, V3C over DASH, live media in XR scenes,...) • Implementation of Reference Application for Android • CMMF (Coded Multisource Media Format) encoder for the xCD-1 code type of ETSI TS 103 973	• FLUTE library supporting MBMS Download, MBS and standard RFC profiles

RELATION WITH SDOS AND INDUSTRY FORA IETF 3GPP ETSI

More information at **5g-mag.com/standards/content-delivery**

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